

3.2.2 All cells arise from other cells

Content	Opportunities for skills development
Within multicellular organisms, not all cells retain the ability to divide. Eukaryotic cells that do retain the ability to divide show a cell cycle. • DNA replication occurs during the interphase of the cell cycle. • Mitosis is the part of the cell cycle in which a eukaryotic cell divides to produce two daughter cells, each with the identical copies of DNA produced by the parent cell during DNA replication. The behaviour of chromosomes during interphase, prophase, metaphase, anaphase and telophase of mitosis. The role of spindle fibres attached to centromeres in the separation of chromatids. Division of the cytoplasm (cytokinesis) usually occurs, producing two new cells. Meiosis is covered in section 3.4.3 Students should be able to: • recognise the stages of the cell cycle: interphase, prophase, metaphase, anaphase and telophase (including cytokinesis) • explain the appearance of cells in each stage of mitosis. Mitosis is a controlled process. Uncontrolled cell division can lead to the formation of tumours and of cancers. Many cancer treatments are directed at controlling the rate of cell division. Binary fission in prokaryotic cells involves: • replication of the circular DNA and of plasmids • division of the cytoplasm to produce two daughter cells, each with a single copy of the circular DNA and a variable number of copies of plasmids. Being non-living, viruses do not undergo cell division. Following injection of their nucleic acid, the infected host cell replicates the virus particles.	
Required practical 2: Preparation of stained squashes of cells from plant root tips; set-up and use of an optical microscope to identify the stages of mitosis in these stained squashes and calculation of a mitotic index. Students should measure the apparent size of cells in the root tip and calculate their actual size using the formula: $\text{Actual size} = \frac{\text{size of image}}{\text{magnification}}$	AT d and e MS 0.3 Calculation of a mitotic index. MS 1.8